

Number Pyramid and Floyd's Triangle

1. Problem Statement

Develop a **C program** to generate a **Right Number Pyramid** and **Floyd's Triangle** using nested loops. The program should accept the number of rows from the user and display both patterns based on the given number of rows.

2. Input

Number of rows

3. Output

Right Number Pyramid:

```
1
12
123
1234
12345
```

Floyd's Triangle:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

4. Concepts Used

- for loops
- Nested loops
- Pattern printing
- Number sequencing
- Formatted output using printf()

5. Algorithm

1. Start the program.
2. Declare rows, i, j, and num.
3. Read the number of rows.
4. Use nested loops to print the Right Number Pyramid.
5. Reset num to 1.
6. Use nested loops to print Floyd's Triangle and increment num after every number.
7. Display both patterns.
8. Stop the program.

6. Pseudocode

BEGIN

```

INPUT rows

FOR i = 1 TO rows
    FOR j = 1 TO i
        DISPLAY j
    END FOR
    DISPLAY newline
END FOR

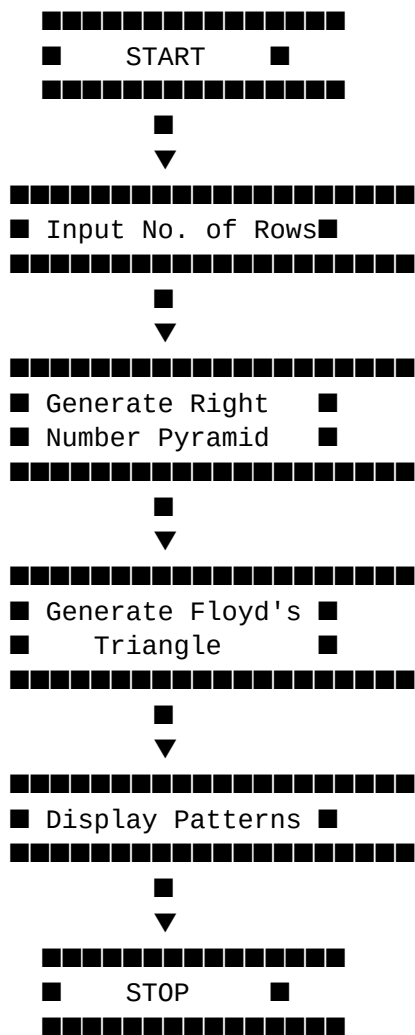
SET num = 1

FOR i = 1 TO rows
    FOR j = 1 TO i
        DISPLAY num
        SET num = num + 1
    END FOR
    DISPLAY newline
END FOR

END

```

7. Block Diagram / Flowchart



8. Sample Test Cases

Test Case	Rows	Expected Result
1	3	Both patterns displayed
2	5	Both patterns displayed
3	7	Both patterns displayed

9. C Program

```
#include <stdio.h>

int main()
{
    int rows, i, j, num = 1;

    printf("Enter number of rows: ");
    scanf("%d", &rows);

    printf("\nRight Number Pyramid:\n");

    for (i = 1; i <= rows; i++)
    {
        for (j = 1; j <= i; j++)
        {
            printf("%d", j);
        }
        printf("\n");
    }

    printf("\nFloyd's Triangle:\n");

    num = 1;

    for (i = 1; i <= rows; i++)
    {
        for (j = 1; j <= i; j++)
        {
            printf("%d ", num);
            num++;
        }
        printf("\n");
    }

    return 0;
}
```

10. Sample Execution

Enter number of rows: 5

Right Number Pyramid:

```
1
12
123
1234
12345
```

Floyd's Triangle:

1
2 3
4 5 6
7 8 9 10
11 12 13 14 15